

TWENTE, Netherlands

WMO: 062900

Lat: 52.2731N

Lon: 6.8908E

Elev: 114

StdP: 14.64

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.1	21.3	10.8	9.6	20.1	15.6	12.2	23.2	25.9	47.5	23.1	46.9	5.2	60	0.546

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.7	85.4	68.0	81.3	66.8	77.6	64.9	70.1	81.5	68.3	78.3	66.5	75.3	6.9	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.5	98.0	74.3	64.7	91.7	72.0	63.1	86.7	70.1	34.1	81.5	32.6	78.3	31.1	75.5	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.6	18.1	16.0	10.9	92.3	6.7	3.6	6.1	94.9	2.2	97.0	-1.6	99.0	-6.5	101.6
			10.2	74.1	6.5	1.6	5.5	75.3	1.7	76.2	-2.0	77.1	-6.7	78.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.0	36.6	37.7	42.1	48.8	55.3	60.3	64.5	63.7	57.8	51.1	43.5	38.1
	DBStd	11.95	8.05	7.49	6.65	7.14	6.92	6.05	5.94	5.53	5.43	6.50	6.50	7.97
	HDD50	1802	416	346	253	107	26	1	0	0	3	67	210	373
	HDD65	5629	880	764	708	486	308	167	83	93	227	433	645	835
	CDD50	1810	1	1	10	71	191	310	449	423	236	100	15	3
	CDD65	165	0	0	0	1	9	26	67	51	10	1	0	0
	CDH74	1691	0	0	0	23	112	297	696	481	79	3	0	0
	CDH80	482	0	0	0	1	18	73	228	149	13	0	0	0
Wind	WSAvg	7.5	9.0	8.6	8.6	7.4	7.1	6.9	6.5	6.2	6.5	7.4	7.6	8.7
Precipitation	PrecAvg	33.2	2.9	2.4	2.2	1.9	2.4	2.8	3.5	3.4	2.9	2.8	3.0	3.1
	PrecMax	43.2	5.2	5.2	5.0	4.1	5.3	5.6	7.7	8.6	5.8	8.5	5.6	7.0
	PrecMin	25.6	0.2	0.9	0.4	0.0	0.9	0.4	1.1	0.7	0.3	0.8	0.3	1.0
	PrecStd	4.5	1.3	1.3	1.3	0.9	1.1	1.3	1.5	1.9	1.4	1.4	1.3	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.6	56.9	65.8	77.4	82.8	87.9	91.6	89.7	81.9	72.1	61.5	56.0
		MCWB	51.0	48.7	53.5	58.6	65.1	69.6	69.0	69.5	67.0	62.9	56.5	52.6
	2%	DB	51.9	52.8	60.2	71.2	76.8	81.1	86.0	84.2	75.7	66.7	57.6	53.6
		MCWB	49.3	48.2	50.9	56.7	61.0	66.2	68.4	67.7	65.1	60.1	54.3	51.1
	5%	DB	49.5	50.6	55.6	66.5	73.1	77.0	81.5	79.4	70.9	63.0	55.2	51.2
		MCWB	47.1	47.2	49.0	54.5	59.7	64.7	67.5	66.6	62.4	58.2	52.5	48.8
	10%	DB	47.0	48.2	52.5	61.9	69.1	72.8	77.2	74.8	67.2	60.4	52.8	48.7
		MCWB	44.8	45.3	47.3	52.6	58.4	62.5	65.5	64.5	60.3	56.7	50.1	46.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.2	51.4	55.3	60.8	67.1	71.3	72.9	71.9	68.6	63.6	57.2	53.6
		MCDB	53.7	54.1	62.0	73.5	77.8	82.4	84.5	84.3	78.0	70.4	58.9	55.0
	2%	WB	49.7	49.3	52.5	57.7	64.0	68.3	70.4	69.5	65.7	60.9	54.8	51.4
		MCDB	51.7	51.6	57.8	67.6	73.6	78.2	82.0	80.5	73.8	65.5	57.3	53.4
	5%	WB	47.2	47.5	50.1	55.7	61.4	66.0	68.6	67.4	63.6	59.0	52.7	49.2
		MCDB	49.1	50.2	54.5	64.1	69.8	74.8	78.9	76.8	69.6	62.3	55.1	51.0
	10%	WB	45.0	45.4	48.1	53.6	59.2	63.7	66.6	65.6	61.6	57.1	50.4	46.8
		MCDB	46.8	48.0	51.9	60.9	66.8	70.9	75.2	73.4	66.2	60.1	52.4	48.6
Mean Daily Temperature Range	5% DB	MDBR	8.6	10.6	14.2	18.7	18.7	19.0	18.7	18.4	16.5	13.6	9.8	8.2
		MCDBR	9.1	13.0	20.5	25.5	24.6	25.3	26.1	25.0	21.3	16.0	11.1	8.6
		MCWBR	8.2	9.6	13.2	13.3	12.2	13.0	11.6	11.5	11.9	10.5	8.8	7.6
	5% WB	MCDBR	9.1	11.1	17.4	22.4	22.0	22.3	23.2	22.5	19.1	15.1	10.7	8.8
		MCWBR	8.6	9.1	12.0	12.5	12.1	12.7	11.2	11.4	11.2	10.4	8.9	8.1
Clear-Sky Solar Irradiance	taub	0.325	0.347	0.376	0.408	0.411	0.409	0.420	0.413	0.389	0.361	0.340	0.321	
	taud	2.356	2.341	2.271	2.222	2.250	2.272	2.249	2.290	2.349	2.405	2.371	2.340	
	Ebn at Noon	211	241	256	261	267	268	262	257	249	233	202	189	
	Edh at Noon	20	26	34	40	41	40	41	37	31	24	19	17	
All-Sky Solar Radiation	RadAvg	217	430	787	1276	1554	1651	1582	1330	967	564	260	169	
	RadStd	26	76	108	138	139	129	200	113	112	54	29	19	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+0.58	N/A	+2.83	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (72 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page